THE PUSHFORWARD ALONG A QUASI-COMPACT QUASI-SEPARATED MORPHISM HAS A RIGHT ADJOINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A9E (sources/stacks-duality/pinned-original-source.tex line 653).

Allowed standard prerequisites (editorial).

- The adjoint functor theorem for compactly generated categories. Derived Categories, Proposition labelled derived-proposition-brown, cited at proof lines 680-682. Supplies the criterion verified in the proof.
- Direct sums and compact generation of the quasi-coherent derived category. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-direct-sums and Theorem labelled perfect-theorem-bondal-van-den-Bergh, cited at proof lines 684-690. Provide the first two hypotheses.
- The pushforward commutes with direct sums for a quasi-compact quasi-separated morphism. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-pushforward-direct-sums, cited at proof lines 692-694. Provides the last hypothesis.

Source reference. Original proposition 0A8G: sources/stacks-chow/tags.txt, source lines 12914–12914. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 664 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DT: sources/stacks-chow/tags.txt, source lines 10648–10648. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 667 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 09IS: sources/stacks-chow/tags.txt, source lines 12050–12050. Cross-chapter standard prerequisite located in the project tag list

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staged in the local snapshot; invoked at proof line 670 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01KV: sources/stacks-chow/tags.txt, source lines 2064–2064. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 672 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 03GI: sources/stacks-chow/tags.txt, source lines 4529–4529. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 673 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DZ: sources/stacks-chow/tags.txt, source lines 10655–10655. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 676 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism between quasi-separated and quasi-compact schemes. The functor $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$ has a right adjoint.*

Proof. We will prove a right adjoint exists by verifying the hypotheses of Derived Categories, Proposition 0A8G. First off, the category $D_{QCoh}(\mathcal{O}_X)$ has direct sums, see Derived Categories of Schemes, Lemma 08DT. The category $D_{QCoh}(\mathcal{O}_X)$ is compactly generated by Derived Categories of Schemes, Theorem 09IS. Since X and Y are quasi-compact and quasi-separated, so is f, see Schemes, Lemmas 01KV and 03GI. Hence the functor Rf_* commutes with direct sums, see Derived Categories of Schemes, Lemma 08DZ. This finishes the proof. □